



Lecture 1 — After class

Introduction to Quantitative Biology | The simplest model of gene expression, and noisy divisions

This sheet contains everything we derived in class, step by step, so that you can redo it alone. Read it with the board photos or your notes next to it. The exercises at the end use only what is here. Exercise 5 is marked **ADVANCED**.

The tags [Python notebook](#) [Section 2b](#) point to the sections of the class notebook (IntroQBio_L1_notebook.ipynb) where the same result is plotted.

1. What is quantitative biology?

In the last decades biology has become very good at making lists of parts: genomes, the set of RNAs in a cell, the set of proteins. A list of parts, however, does not tell us how a cell grows, how it senses its environment, or how it decides what to do. For that we need to understand how the parts interact, and how fast, and how much.

Quantitative biology is **not** biology plus numbers, and it is **not** biology plus computational tools. It uses numbers, models and tools to make **predictions** and to obtain biological **understanding**. Models are useful for three reasons:

- a model forces us to propose a *mechanism* that explains the data;
- a model makes *predictions* that an experiment can test — and when the prediction fails, we learn which assumption was wrong;
- a model *simplifies*: it leaves out details on purpose, so that one aspect of reality becomes visible.

KEY TAKEAWAY

The modelling loop, which we follow in every lecture: (1) ask a biological question → (2) build a model and state its assumptions → (3) solve it, choosing the right tool → (4) interpret the result and compare it with numbers → (5) refine the model.

Some numbers of an *E. coli* cell (Alon, Tables 1.1 and 1.2)

About 4×10^6 proteins of about 4500 types, so on average about 900 copies of each type (many proteins, for example transcription factors, have far fewer). One protein per cell corresponds to a concentration of about 1 nM. Making a protein takes about a minute; a typical protein lives about an hour; a generation lasts from 20 minutes to several hours. These timescales differ by orders of magnitude, which is what allows us to neglect the fastest steps in the model below.

2. The simplest model of gene expression

The model and its assumptions

A gene is switched on. Its protein has concentration $Y(t)$. The protein is **produced** at a constant rate β and **removed** at a rate αY : each protein has the same probability per unit time α of being removed, so

the more proteins there are, the faster they are removed. Removal has two parts: active degradation, and dilution because the cell grows. Following Alon [1]:

$$\frac{dY}{dt} = \beta - \alpha Y, \quad \alpha = \alpha_{\text{dil}} + \alpha_{\text{deg}}.$$

β is a concentration per unit time; α is in units of 1/time.

Assumptions we made

- The cell is well mixed: one number Y describes it.
- β and α are constant in time, and production does not depend on Y .
- The minutes needed for transcription and translation are neglected: they are short compared with the time for the protein level to change (about one hour).
- Y is a continuous, deterministic quantity. Sections 3 and 4 show where this stops being enough.

Reading the equation

If $Y < \beta/\alpha$, then $dY/dt > 0$ and Y increases. If $Y > \beta/\alpha$, it decreases. At

$$Y_{\text{st}} = \frac{\beta}{\alpha}$$

production and removal balance and Y stops changing: this is the **steady state**. From both sides Y moves towards it, so it is stable.

Solving it: method 1, separation of variables

This is the method of the sheet before class, applied to the general $dx/dt = f(x)$ — the question you were left with. Separate the variables, that is, put everything that contains x on one side and everything that contains t on the other:

$$\frac{dx}{dt} = f(x) \implies \frac{dx}{f(x)} = dt.$$

Now integrate both sides over the same stretch of time: the right-hand side from 0 to t , the left-hand side between the values of x reached at those two instants, x_0 and $x(t)$:

$$\int_{x_0}^{x(t)} \frac{dx'}{f(x')} = \int_0^t dt' = t.$$

(We call the integration variables x' and t' to keep them distinct from the limits $x(t)$ and t .) This is as far as one can go for a general f , and notice what it gives: not $x(t)$, but t as a function of x . Two things can still stop us — computing the integral, and inverting the result to isolate x . When either is impossible we solve the equation numerically (Lecture 2).

For our equation both steps can be done. With $f(Y) = \beta - \alpha Y$:

$$\int_{Y_0}^Y \frac{dY'}{\beta - \alpha Y'} = t.$$

Step 1 — the integral. Substitute $u = \beta - \alpha Y'$, so that $du = -\alpha dY'$, i.e. $dY' = -du/\alpha$. The limits move with the variable: $Y' = Y_0$ gives $u_0 = \beta - \alpha Y_0$, and $Y' = Y$ gives $u = \beta - \alpha Y$. Then

$$\int_{Y_0}^Y \frac{dY'}{\beta - \alpha Y'} = -\frac{1}{\alpha} \int_{u_0}^u \frac{du'}{u'} = -\frac{1}{\alpha} [\ln u']_{u_0}^u = -\frac{1}{\alpha} \ln \frac{\beta - \alpha Y}{\beta - \alpha Y_0}.$$

The two logarithms have been combined into the logarithm of the ratio, which is what makes the next step easy. (If $Y_0 < \beta/\alpha$ the protein never crosses the steady state, so $\beta - \alpha Y'$ stays positive all along the path and the logarithm is well defined; the same is true from above, and in both cases the ratio is positive.)

Step 2 — invert it. Set the result equal to t and multiply both sides by $-\alpha$:

$$-\frac{1}{\alpha} \ln \frac{\beta - \alpha Y}{\beta - \alpha Y_0} = t \implies \ln \frac{\beta - \alpha Y}{\beta - \alpha Y_0} = -\alpha t.$$

Take the exponential of both sides, which removes the logarithm:

$$\frac{\beta - \alpha Y}{\beta - \alpha Y_0} = e^{-\alpha t} \implies \beta - \alpha Y = (\beta - \alpha Y_0) e^{-\alpha t}.$$

Step 3 — isolate Y . Move αY to one side and divide by α :

$$\alpha Y = \beta - (\beta - \alpha Y_0) e^{-\alpha t} \implies Y(t) = \frac{\beta}{\alpha} - \left(\frac{\beta}{\alpha} - Y_0 \right) e^{-\alpha t} = \frac{\beta}{\alpha} + \left(Y_0 - \frac{\beta}{\alpha} \right) e^{-\alpha t},$$

which is the solution boxed below. The last equality is only a change of sign inside the bracket, written that way because $Y_0 - \beta/\alpha$ is the distance from the starting point to the steady state.

Solving it: method 2, homogeneous + particular solution

This method works for every **linear** ODE (an equation where Y and dY/dt appear only to the first power). It never requires computing an integral, and it is the one we will keep using in the rest of the course.

Step 1 — the homogeneous equation. Remove the term that does not contain Y , here the constant β :

$$\frac{dY_h}{dt} = -\alpha Y_h.$$

This is the equation of the sheet before class with $k = \alpha$: either guess $Y_h = C e^{bt}$, whose derivative is $b Y_h$, so that $b = -\alpha$; or separate the variables, $dY_h/Y_h = -\alpha dt$, and integrate. Either way

$$Y_h(t) = C e^{-\alpha t}, \quad C \text{ any constant.}$$

The constant stays undetermined on purpose: we will fix it at the very end.

Step 2 — one particular solution. We now need *any single* function that solves the full equation, however we find it — there is no need for it to be general. The simplest thing to try is a constant, Y_p , and it is a reasonable guess because the equation says that Y stops changing somewhere. A constant has zero derivative, so the equation becomes

$$0 = \beta - \alpha Y_p \implies Y_p = \frac{\beta}{\alpha} = Y_{st} :$$

the particular solution is the steady state we had already read off the equation.

Step 3 — add them. Claim: $Y = Y_h + Y_p$ solves the full equation. Check it by computing the two sides separately. Left-hand side:

$$\frac{d}{dt}(Y_h + Y_p) = \frac{dY_h}{dt} + 0 = -\alpha Y_h.$$

Right-hand side:

$$\beta - \alpha(Y_h + Y_p) = \underbrace{\beta - \alpha Y_p}_{= 0 \text{ by Step 2}} - \alpha Y_h = -\alpha Y_h.$$

They agree, so the sum is a solution. This works because the equation is linear: the constant β is taken care of by Y_p , and what is left over is exactly the homogeneous equation that Y_h solves.

Step 4 — the initial condition. So far C is free, and indeed we have not used the initial condition yet. We have one unknown constant and one condition, so C is fixed by

$$Y(0) = C e^0 + \frac{\beta}{\alpha} = C + \frac{\beta}{\alpha} = Y_0 \implies C = Y_0 - \frac{\beta}{\alpha}.$$

Putting it back into $Y = Y_h + Y_p$, both methods give

$$Y(t) = \frac{\beta}{\alpha} + \left(Y_0 - \frac{\beta}{\alpha}\right) e^{-\alpha t}$$

Always check a solution you have just derived. Three cheap checks:

- at $t = 0$, $e^0 = 1$ and the bracket cancels the β/α , leaving $Y(0) = Y_0$;
- as $t \rightarrow \infty$, $e^{-\alpha t} \rightarrow 0$ and $Y \rightarrow \beta/\alpha = Y_{\text{st}}$;
- differentiating, $dY/dt = -\alpha(Y_0 - \beta/\alpha)e^{-\alpha t}$, while $\beta - \alpha Y(t) = \beta - \beta - \alpha(Y_0 - \beta/\alpha)e^{-\alpha t}$ gives the same thing — so the solution does satisfy the equation.

Which method should you use?

Separation of variables works for *any* $dx/dt = f(x)$, linear or not, but it asks for an integral and an inversion, and either can be impossible. Homogeneous + particular works only for linear equations, but when it works it is faster and needs no integral. The second is the one that will generalise later in the course; the first is the one to fall back on when the equation is not linear.

[Python notebook](#) [Section 2a](#)

Response time

Take a gene that is switched on at $t = 0$ in a cell that had none of its protein, so $Y_0 = 0$. Putting $Y_0 = 0$ in the boxed solution and factoring out β/α :

$$Y(t) = \frac{\beta}{\alpha} - \frac{\beta}{\alpha} e^{-\alpha t} = Y_{\text{st}} (1 - e^{-\alpha t}).$$

The **response time** $T_{1/2}$ is the time needed to reach half of the steady state, $Y(T_{1/2}) = Y_{\text{st}}/2$. Divide both sides by Y_{st} — which already removes β from the problem — and solve:

$$1 - e^{-\alpha T_{1/2}} = \frac{1}{2} \implies e^{-\alpha T_{1/2}} = \frac{1}{2} \implies -\alpha T_{1/2} = \ln \frac{1}{2} = -\ln 2 \implies T_{1/2} = \frac{\ln 2}{\alpha}.$$

Read the chain backwards and you see why β cannot appear in the answer: it was divided away in the first step.

[Python notebook](#) [Section 2b](#)

KEY TAKEAWAY

The production rate β sets how much protein there is at steady state. Only the removal rate α sets how fast the steady state is reached.

Stable proteins respond in one generation

Many proteins are not actively degraded ($\alpha_{\text{deg}} = 0$), so only dilution removes them. If production stops, after one cell generation τ the volume has doubled and the concentration has halved. So $T_{1/2} = \tau$, and $\alpha_{\text{dil}} = \ln 2/\tau$ (Alon, Section 1.4.1). For an *E. coli* dividing every 30 min, $\alpha = 0.023 \text{ min}^{-1}$ and the

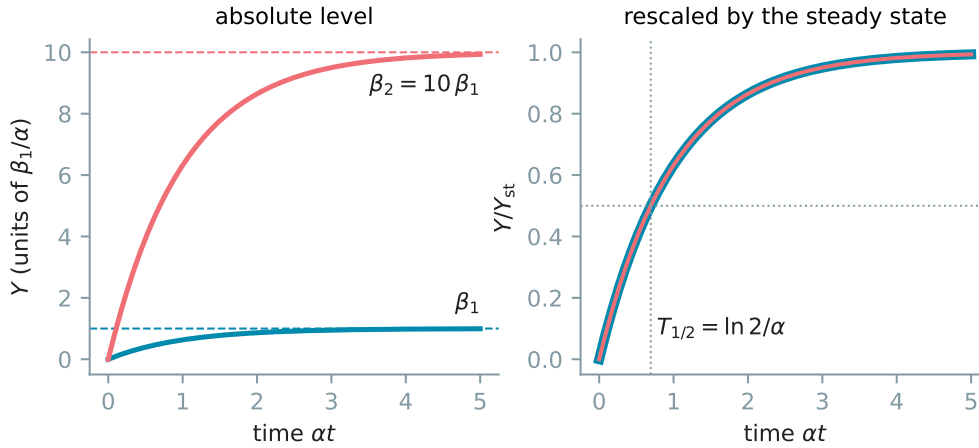


Figure 1: Two genes with the same α ; the second is produced ten times faster. Left: it reaches a ten times higher level. Right: divided by their own steady states, the two curves are the same, and both reach one half at $T_{1/2} = \ln 2/\alpha$. Time is in units of $1/\alpha$.

response takes 30 min.

A cell can respond faster by degrading the protein actively (larger α). The price: to keep the same steady state β/α , it must produce more (β must grow with α).

[Python notebook](#) [Section 2c](#)

3. Switching off production

Question. A stable protein is at steady state. At $t = 0$ its gene is switched off. How many generations do we have to wait until the protein has disappeared?

With $\beta = 0$ the equation is the homogeneous one, $dY/dt = -\alpha Y$, and with $Y(0) = Y_{st}$:

$$Y(t) = Y_{st} e^{-\alpha t}.$$

For a stable protein $\alpha = \ln 2/\tau$. After n generations ($t = n\tau$), using $e^{-n \ln 2} = 2^{-n}$:

$$Y(n\tau) = \frac{Y_{st}}{2^n}.$$

The protein is halved at every generation. As time goes on, Y goes to zero, but it never becomes exactly zero in a finite time.

Counting molecules. Y is also a number of molecules per cell, and molecules come one by one: a cell cannot contain half a protein. So the interesting question is not when Y reaches zero, but when it drops below one molecule per cell. That happens when

$$\frac{Y_{st}}{2^n} < 1 \iff 2^n > Y_{st} \iff n > \log_2 Y_{st},$$

where the last step takes the logarithm in base 2 of both sides (allowed, and the inequality keeps its direction, because $\log_2$ is an increasing function). So the protein is practically gone after about

$$n = \log_2 Y_{st} \text{ generations.}$$

For $Y_{st} = 1000$ molecules this is about 10 generations (5 hours if $\tau = 30$ min); for 100 molecules 6.6 generations; for 10 molecules 3.3 generations.

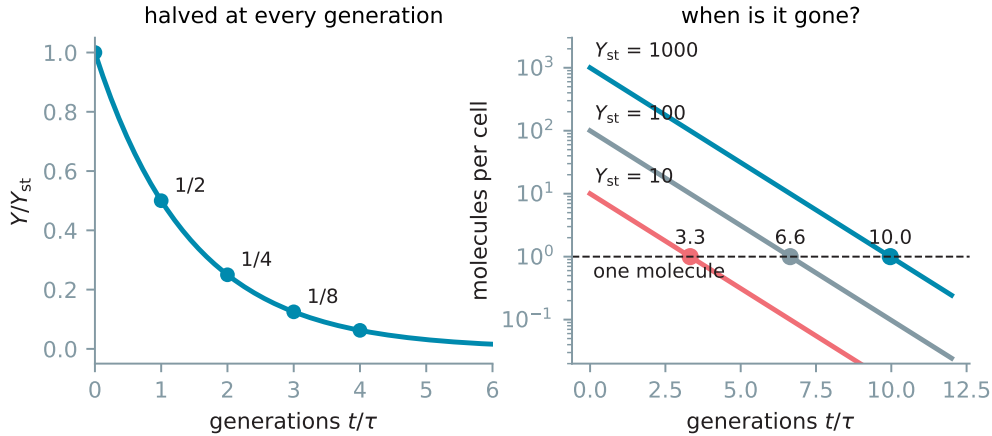


Figure 2: Production switched off at $t = 0$ for a stable protein. Left: the level is halved at every generation. Right: the number of molecules per cell, on a logarithmic axis, for $Y_{st} = 10, 100, 1000$. The dashed line is one molecule; it is reached after $\log_2 Y_{st}$ generations (3.3, 6.6 and 10.0).

[Python notebook](#) [Section 3](#)

Is it the same in every cell? The equation says that every cell has exactly $Y_{st}/2^n$ molecules after n generations. Halving, however, is what happens *on average*: at each division the molecules are shared at random between the two daughters. Section 4 computes how large the difference between two sister cells is.

4. Partitioning at cell division

When a cell divides, how are its N molecules shared between the two daughters? This section follows Phillips *et al.* [2], Chapter 2, pp. 43–48.

Assumptions

The two daughters have the same volume. Each molecule ends up in daughter 1 with probability $1/2$, independently of the others. No molecule is actively sent to one side.

Each molecule flips a coin

With these assumptions, sharing N molecules is exactly tossing N fair coins and counting the heads. In the bootcamp, the probability of k heads in N tosses, each with probability p of heads, was the **binomial distribution**

$$P(k) = \binom{N}{k} p^k (1-p)^{N-k}, \quad \langle k \rangle = Np, \quad \text{Var}(k) = Np(1-p).$$

Here the “heads” are the molecules that go to daughter 1: $k = n_1$ and $p = 1/2$. Since $p^{n_1}(1-p)^{N-n_1} = (1/2)^{n_1}(1/2)^{N-n_1} = 1/2^N$, daughter 1 receives n_1 molecules (and daughter 2 receives $n_2 = N - n_1$) with probability

$$P(n_1) = \binom{N}{n_1} \frac{1}{2^N}, \quad \langle n_1 \rangle = \frac{N}{2}, \quad \text{Var}(n_1) = \frac{N}{4}.$$

The standard deviation is $\sigma_{n_1} = \sqrt{N}/2$, so the **relative noise** is

$$\frac{\sigma_{n_1}}{\langle n_1 \rangle} = \frac{\sqrt{N}/2}{N/2} = \frac{1}{\sqrt{N}}.$$

It is 0.32 for $N = 10$, 0.10 for $N = 100$ and 0.03 for $N = 1000$. For $N = 10$, daughter 1 receives exactly 5 molecules with probability $252/1024 \approx 0.25$ only.

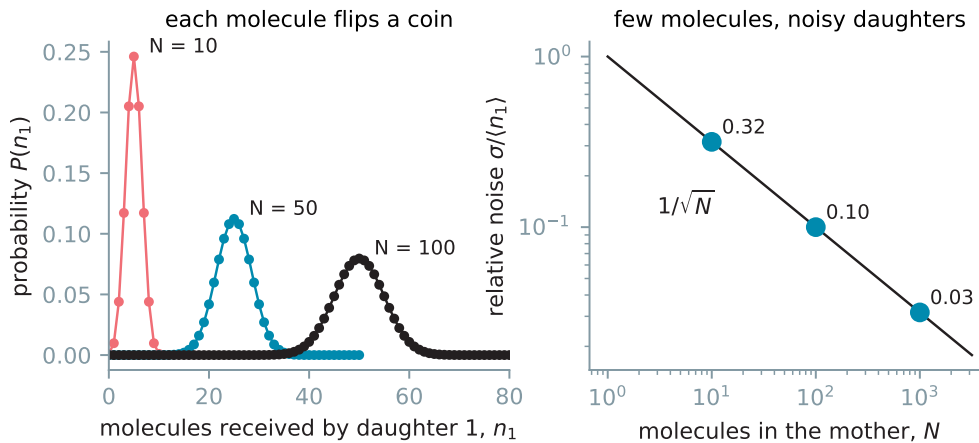


Figure 3: Left: probability that daughter 1 receives n_1 molecules, for $N = 10, 50, 100$. Right: relative noise $1/\sqrt{N}$, with the values for $N = 10, 100$ and 1000 .

[Python notebook](#) [Section 4](#)

KEY TAKEAWAY

Two genetically identical daughters of the same mother can differ just because molecules are few and the sharing is random. The fewer the molecules, the larger the relative difference.

Counting molecules with a microscope

Under the microscope we do not see molecules: we see light. A cell containing n fluorescent proteins appears with an intensity I , measured in whatever units the camera produces (counts, grey levels, arbitrary units). To go from the picture back to biology we need to know how many molecules make up that light. The problem is that nobody has told us how bright *one* molecule is.

The model of the measurement

We assume the intensity is **proportional** to the number of molecules,

$$I = \nu n,$$

with the same constant ν (intensity per molecule) for every molecule and every cell, and with the background of the image already subtracted, so that $n = 0$ gives $I = 0$. The constant ν is unknown: it depends on the protein, on the microscope, on the exposure time.

At a division we can measure the two daughters and the mother:

$$I_1 = \nu n_1, \quad I_2 = \nu (N - n_1), \quad I_{\text{tot}} = I_1 + I_2 = \nu n_1 + \nu (N - n_1) = \nu N.$$

So I_1 , I_2 and I_{tot} are numbers we read off the images, while N , n_1 and ν are all hidden from us. The idea of Rosenfeld *et al.* is that the *randomness* of the partitioning is enough to reveal ν . To use it we first need to know how means and variances behave when a random number is multiplied by a constant — here, by ν .

Mean and variance of $aX + c$

First, the notation. The angular brackets $\langle \cdot \rangle$ always mean *the average of whatever is written inside them*, taken over all the possible outcomes, each weighted by its probability. So $\langle n_1 \rangle$ is the average number of molecules received by daughter 1, $\langle n_1^2 \rangle$ is the average of the *square* of that number, and $\langle (I_1 - I_2)^2 \rangle$ is the average of the squared difference over many divisions. The brackets are not a function applied to a number: what is inside must be a random quantity, and what comes out is an ordinary number.

Let X be a random number that takes the value x with probability $P(x)$. Its **mean** and its **variance** are the two sums

$$\langle X \rangle = \sum_x x P(x), \quad \text{Var}(X) = \langle (X - \langle X \rangle)^2 \rangle = \sum_x (x - \langle X \rangle)^2 P(x).$$

The variance is the average *squared* distance from the mean: it is never negative, and it has the units of X squared (if X is a number of molecules, $\text{Var}(X)$ is in molecules²). The square root $\sigma_X = \sqrt{\text{Var}(X)}$, the standard deviation, is back in the units of X .

Now take two *constants* a and c (fixed numbers, not random) and build the new random number $Y = aX + c$. It takes the value $ax + c$ exactly when X takes the value x , so it takes it with the same probability $P(x)$. Everything follows from that one sentence.

1. The mean is linear.

$$\langle aX + c \rangle = \sum_x (ax + c) P(x) = a \sum_x x P(x) + c \sum_x P(x) = a \langle X \rangle + c,$$

because $\sum_x P(x) = 1$. Multiplying by a multiplies the mean by a ; adding c shifts the mean by c .

2. The variance ignores c and picks up a^2 . Subtract the mean of Y from Y :

$$Y - \langle Y \rangle = (aX + c) - (a\langle X \rangle + c) = a(X - \langle X \rangle).$$

The constant c cancels — shifting every value by the same amount moves the mean by the same amount and changes no distance. Squaring and averaging,

$$\text{Var}(aX + c) = \langle a^2(X - \langle X \rangle)^2 \rangle = a^2 \text{Var}(X), \quad \text{so} \quad \sigma_{aX+c} = |a| \sigma_X.$$

The two special cases are the ones we use: $\text{Var}(aX) = a^2 \text{Var}(X)$ and $\text{Var}(X + c) = \text{Var}(X)$. The factor is a^2 , *not* a , because the variance is built from squares; the standard deviation is the one that scales like a .

3. The square of the mean is not the mean of the square. Expanding the definition,

$$\text{Var}(X) = \langle X^2 - 2X\langle X \rangle + \langle X \rangle^2 \rangle = \langle X^2 \rangle - 2\langle X \rangle \langle X \rangle + \langle X \rangle^2 = \langle X^2 \rangle - \langle X \rangle^2,$$

where we used that $\langle X \rangle$ is a constant. Therefore

$$\langle X^2 \rangle = \text{Var}(X) + \langle X \rangle^2, \quad \text{and in particular} \quad \langle X \rangle = 0 \implies \langle X^2 \rangle = \text{Var}(X).$$

A check you can do by hand

Take $N = 2$ molecules, so $n_1 = 0, 1, 2$ with probabilities $1/4, 1/2, 1/4$. Directly from the definitions: $\langle n_1 \rangle = 0 \cdot \frac{1}{4} + 1 \cdot \frac{1}{2} + 2 \cdot \frac{1}{4} = 1 = N/2$, and $\langle n_1^2 \rangle = 0 \cdot \frac{1}{4} + 1 \cdot \frac{1}{2} + 4 \cdot \frac{1}{4} = \frac{3}{2}$, so $\text{Var}(n_1) = \frac{3}{2} - 1 = \frac{1}{2} = N/4$. ✓

Now build $Y = 2n_1 - N = 2n_1 - 2$, which takes the values $-2, 0, 2$ with probabilities $1/4, 1/2, 1/4$. Directly: $\langle Y \rangle = 0$ and $\langle Y^2 \rangle = 4 \cdot \frac{1}{4} + 0 \cdot \frac{1}{2} + 4 \cdot \frac{1}{4} = 2$. The rules give $\langle Y \rangle = 2\langle n_1 \rangle - 2 = 0$ and $\langle Y^2 \rangle = \text{Var}(Y) = 2^2 \text{Var}(n_1) = 4 \cdot \frac{1}{2} = 2$. ✓

Note that $2^2 \text{Var}(n_1) = 2$ while $2 \text{Var}(n_1) = 1$: forgetting the square would have cost a factor 2 here, and a factor a^2/a in general.

The derivation

Step 1 — the difference between the two daughters.

$$I_1 - I_2 = \nu n_1 - \nu(N - n_1) = \nu(2n_1 - N).$$

In this expression ν is an unknown constant, N is fixed for a given mother cell, and n_1 is the only random quantity: everything is of the form $aX + c$ with $X = n_1$, $a = 2\nu$ and $c = -\nu N$.

Step 2 — why we square it. By rule 1, the mean of the difference is

$$\langle I_1 - I_2 \rangle = 2\nu \langle n_1 \rangle - \nu N = 2\nu \frac{N}{2} - \nu N = 0.$$

This had to happen: which daughter we call “1” is our arbitrary choice, so positive and negative differences are equally likely and they cancel. Averaging $I_1 - I_2$ over many divisions therefore tells us nothing. What does not cancel is the *squared* difference, which measures how far apart the two daughters are, irrespective of sign. And because the mean is zero, rule 3 says the mean square *is* the variance:

$$\langle (I_1 - I_2)^2 \rangle = \text{Var}(I_1 - I_2).$$

Step 3 — apply the rules. With $a = 2\nu$ and $c = -\nu N$, rule 2 gives

$$\text{Var}(I_1 - I_2) = \text{Var}(2\nu n_1 - \nu N) = (2\nu)^2 \text{Var}(n_1) = 4\nu^2 \cdot \frac{N}{4} = \nu^2 N.$$

Step 4 — eliminate what we cannot measure. The result $\nu^2 N$ contains the two unknowns ν and N . But they appear in the measurable combination $I_{\text{tot}} = \nu N$, and we can split $\nu^2 N = \nu \cdot (\nu N)$. Hence

$$\langle (I_1 - I_2)^2 \rangle = \nu I_{\text{tot}}$$

The hidden molecule count N has disappeared. On the left there is a quantity we can measure (the squared difference between two daughters), on the right a quantity we can measure (I_{tot}) multiplied by the single unknown ν .

Another route to the same result

There is another route to the boxed result, which does not use the rules for $\text{Var}(aX + c)$ at all: it needs only $\langle n_1 \rangle = N/2$ and $\langle n_1^2 \rangle = \text{Var}(n_1) + \langle n_1 \rangle^2$. Expand the square first:

$$(I_1 - I_2)^2 = \nu^2(2n_1 - N)^2 = \nu^2(4n_1^2 - 4Nn_1 + N^2),$$

then average term by term, using that the average of a sum is the sum of the averages:

$$\langle (2n_1 - N)^2 \rangle = 4 \langle n_1^2 \rangle - 4N \langle n_1 \rangle + N^2.$$

With $\langle n_1 \rangle = N/2$ and $\text{Var}(n_1) = N/4$,

$$\langle n_1^2 \rangle = \frac{N}{4} + \left(\frac{N}{2}\right)^2 = \frac{N}{4} + \frac{N^2}{4},$$

so

$$\langle (2n_1 - N)^2 \rangle = 4 \left(\frac{N}{4} + \frac{N^2}{4} \right) - 4N \frac{N}{2} + N^2 = N + N^2 - 2N^2 + N^2 = N,$$

and $\langle (I_1 - I_2)^2 \rangle = \nu^2 N = \nu I_{\text{tot}}$ as before. It is longer, but it shows where the N that survives comes from: the N^2 terms cancel exactly, and what is left is the variance of the partitioning.

What it is good for

Every division gives one pair of numbers: I_{tot} (how bright the mother was) and $(I_1 - I_2)^2$ (how unequal the sharing was). Different mothers have different N , hence different I_{tot} , so a movie of many divisions covers a range of I_{tot} . Group the divisions into bins of similar I_{tot} , average $(I_1 - I_2)^2$ inside each bin, and plot that average against I_{tot} . The boxed result says the points lie on a **straight line through the origin**, and that its **slope is ν** , the intensity of a single molecule.

Once ν is known, the calibration is done and any cell can be converted from light to molecules:

$$n = \frac{I}{\nu}.$$

For instance, if the fit gives a slope $\nu = 120$ (intensity units per molecule), a cell whose intensity is $I = 3600$ contains $n = 3600/120 = 30$ molecules. Check the units: the slope of $\langle (I_1 - I_2)^2 \rangle$ (intensity²) against I_{tot} (intensity) is indeed an intensity, and by construction it is the intensity of one molecule.

The plot also tests the model, which is why it is worth making rather than assuming. If the points fall on a straight line through the origin, random binomial partitioning describes the data. A curve, or a line that misses the origin, says that something else contributes: unequal division (Exercise 5), active segregation, or a measurement noise of its own.

KEY TAKEAWAY

The randomness we could have called a nuisance is what calibrates the microscope: because the fluctuations of a binomial are set by the number of molecules, measuring the fluctuations measures the number. Rosenfeld *et al.* (*Science*, 2005) used exactly this on real data (see the slides).

Exercises

Exercise 1: Changing the production rate

A gene is at steady state with production rate β_1 . At $t = 0$ the production rate suddenly becomes β_2 (removal rate α unchanged).

- What is $Y(0)$? Use the boxed solution of Section 2 to write $Y(t)$ for $t > 0$.
- Sketch $Y(t)$ for $\beta_2 > \beta_1$ and for $\beta_2 < \beta_1$. What is the case $\beta_2 = 0$?
- Show that the time to go halfway from the old to the new steady state is $\ln 2/\alpha$ in both cases.

You can compare your answer with Alon, Exercise 1.1.

Exercise 2: Putting numbers on it

An *E. coli* cell divides every 60 min. Protein A is stable. Protein B is also actively degraded, with a degradation half-life of 20 min (the time it would take to lose half of it by degradation alone).

- Compute α_{dil} and, for protein B, α_{deg} (in min^{-1}).
- Compute the response time $T_{1/2}$ of A and of B.
- A and B have the same steady-state concentration. How many times larger must β_B be than β_A ?
- Both proteins are at steady state with 1000 molecules per cell when their genes are

switched off. After how many minutes is there less than one molecule per cell, for A and for B? How many generations is that?

Exercise 3: One curve for all genes

Define a time without units $s = \alpha t$ and a level without units $y = Y/Y_{\text{st}}$.

- Using $dY/dt = \alpha Y_{\text{st}} dy/ds$, show that the equation becomes $dy/ds = 1 - y$.
- How many parameters are left? Use this to explain the right panel of Figure 1.

Exercise 4: Partitioning with numbers

- A mother cell has $N = 100$ molecules. What are the mean and the standard deviation of n_1 ? Same question for $N = 10$.
- For $N = 10$, what is the probability that daughter 1 receives no molecule? That one of the two daughters (either) receives none?
- A student writes $\text{Var}(2n_1 - N) = 2 \text{Var}(n_1)$. Which of the three rules is being misused, and by what factor is the resulting $\langle (I_1 - I_2)^2 \rangle$ wrong? Check the correct value for $N = 4$ by hand: list the five possible values of $2n_1 - N$ with their probabilities and average the square directly.
- In an (invented) experiment, the points $\langle (I_1 - I_2)^2 \rangle$ against I_{tot} lie on a line of slope 250 (intensity units). A cell has intensity 5000. About how many molecules does it contain?
- Instead of the squared difference, some people plot the *relative* difference $(I_1 - I_2)/I_{\text{tot}}$. For a given mother, I_{tot} is a constant, so the rules apply again. Show that $\langle ((I_1 - I_2)/I_{\text{tot}})^2 \rangle = 1/N = \nu/I_{\text{tot}}$, and say what you would plot against what to read ν off the slope.

Exercise 5: ADVANCED — Unequal division

Suppose daughter 1 is larger, and each molecule goes to it with probability $p \neq 1/2$.

- Give $\langle n_1 \rangle$ and $\text{Var}(n_1)$.
- Compute the mean of $2n_1 - N$. Is it still 0?
- Using $\langle X^2 \rangle = \text{Var}(X) + \langle X \rangle^2$, show that

$$\langle (I_1 - I_2)^2 \rangle = \nu^2 [N^2(2p - 1)^2 + 4Np(1 - p)].$$

- Is $\langle (I_1 - I_2)^2 \rangle$ still proportional to I_{tot} ? What would you see on the plot if you assumed equal division but the cells divided unequally?

References

- [1] Uri Alon. *An Introduction to Systems Biology: Design Principles of Biological Circuits*. 2nd. CRC Press, 2019.
- [2] Rob Phillips et al. *Physical Biology of the Cell*. 2nd. Garland Science, 2013.